Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous Institute Affiliated to University of Mumbai)

MSE Examination

Max. Marks: 30 October 2023 Duration: 60 min.
Class: T.E. Semester-V Computer Engineering Course Code: CS301
Name of the Course: Software Engineering

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO
Q.1 A)	Illustrate spiral model with diagram. List two advantages and two disadvantages of the spiral model.	5	CO1
Q.1 B)	Suggest suitable life cycle model with reasons for the following i. A software for an academic institution to automate its: Course registration and grading, Fee collection, Staff salary, Purchase and store inventory. The software would be developed by tailoring similar software that was developed for another educational institution: 70% reuse, 10% new code and 20% modification. ii. A mobile phone vendor: A new mobile operating system by upgrading the existing operating system Needs to work well efficiently with 4G systems, Power usage minimization, Directly upload backup data on a cloud infrastructure maintained by the mobile phone vendor.	5	CO1
Q.2 A)	What is the difference between requirement analysis and specification? What are the important activities carried out during requirement analysis and specification phase?	5	CO1
Q.2 B)	Identify functional and non-functional requirements of a open source learning platform (moodle.org) system	5	CO1
Q.3 A)	A student wants to issue a book from Library. At first the student requests to the librarian for the book which he/she needs. The librarian searches the requested book in the database. The book	5	CO2

	database system searches the book availability and checks the details of the book. The database system returns the details of the book to the librarian. Then the librarian requests library card from the student. The student provides the library card and the librarian updates the details into database. After completion of entering the details in the database, a receipt is generated for the librarian. Finally, the book is issued to the student. Based on the details of the above scenario draw the sequence diagram.		
Q.3 B)	Draw a state diagram for the online order delivery system. You must include the following: i. The states that must be included in the diagram are: (Checking, dispatching, waiting, cancelled, and delivered. (Additionally you may include more states) ii. Check whether all the items are available or not. iii. Include whether all the items are checked or not. iv. Include if some items are not in the stock. For (ii) – (iv), show appropriate transitions from a given state. Use the concept of *Superstate*.	5	CO2